

Article

CORES: Context-Aware Emotion-Driven Recommendation System-Based LLM to Improve Virtual Shopping Experiences

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Abstract: In today's business landscape, artificial intelligence (AI) plays a pivotal role in shopping processes and customization. As the demand for customization grows, virtual reality (VR) emerges as an innovative solution to improve users' perception and decision making in virtual shopping experiences (VSEs). Despite its potential, limited research has explored the integration of contextual information and emotions in VR to deliver effective product recommendations. This paper presents CORES (context-aware emotion-driven recommendation system), a novel approach designed to enrich users' experiences and to support decision making in VR. CORES combines advanced large language models (LLMs) and embedding-based context-aware recommendation strategies to provide customized products. Therefore, emotions are collected from social platforms, and relevant contextual information is matched to enable effective recommendation. Additionally, CORES leverages transformers and retrieval-augmented generation (RAG) capabilities to explain recommended items, facilitate VR visualization, and generate insights using various prompt templates. CORES is applied to a VR shop of different items. An empirical study validates the efficiency and accuracy of this approach, achieving a significant average accuracy of 97% and an acceptable response time of 0.3267s in dynamic shopping scenarios.

Keywords: virtual reality; emotions; recommendation; LLM; e-commerce; context



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1. Introduction

Global e-commerce is evolving rapidly, covering a broader range of online shopping activities and intensifying market competition. To thrive in this field, companies must harness modern artificial intelligence (AI) technologies to engage customers, uncover valuable insights, and enhance customer experience. The widespread adoption of AI-powered recommender systems highlights their crucial role in modern online shopping. By providing relevant products suggestions, AI solutions enhance engagement and the shopping experience.

The main recommender systems used in the context of online shopping are content-based [1], collaborative filtering [2], deep learning [3], hybrid, and context-aware recommendation [4]. With the increasing variety of available products, clients seek customized products that align with their intentions, desires, and emotions. In response, online shopping is shifting toward virtual reality (VR) technology, with the introduction of immersive virtual shopping experiences (VSEs) [5] through 3D product models. VR enables users to explore products in a simulated real-world environment, providing a deeper understanding of their features.

Several studies have highlighted the impact of VR avatars on users' decision-making behavior [6], offering valuable insights into products and purchasing processes. Despite its benefits, VR-based online shopping often lacks contextual cues such as shopping context, delivery time, users' intentions, positive emotion responses towards products, and past experiences or interactions. These factors are essential for delivering effective recommendations. In the rapidly evolving retail landscape, technology has revolutionized how consumers interact with products and brands. This case study explores VRShop, a VR shopping platform that enables users to experience immersive shopping across various stores. The platform features an advanced recommendation system that enhances users' experiences by suggesting relevant products based on their preferences and behaviors. VRShop was developed to meet the growing demand for innovative online shopping experiences. VRShop bridges the gap between traditional e-commerce platforms and physical stores, combining convenience with engaging realistic interactions.

With the emergence of the large language model (LLM) [7], personalized item recommendation can be effectively enhanced, as LLMs excel in generative capabilities and information integration abilities [8,9]. However, LLMs face challenges such as insufficient domain-specific knowledge, which can result in inaccuracies or incomplete responses. These challenges become apparent when processing diverse inputs including social media data, large product catalogs, and varied user feedback. Retrieval-augmented generation (RAG) offers a solution to these challenges by retrieving external contextually enriched data and incorporating them into responses [10]. For instance, Wang et al. [11] proposed a framework that uses structured knowledge bases to enhance RAG in dialogue systems. RAG has also proven effective in domains like healthcare, education, and law [12–14], supporting robust data retrieval and content generation. However, these approaches lack contextual sensitivity to personal emotions, individual preferences, and behavioral tendencies. Furthermore, they have not been effectively adapted to practical use within VR environments, limiting their ability to deliver deeply personalized and immersive recommendations.

To address these shortcomings, we introduce CORES, a novel context-aware emotion-driven recommendation system for VR shopping experiences, which integrates users' profiles and contextual emotions to improve recommendations. CORES leverages the power of LLM by amalgamating VR context, users' emotional states, and dataset inputs to recommend relevant and personalized items. The system also uses 3D reconstruction techniques to visualize recommended items in a VR environment. By integrating emotional context and pre-trained transformer models, CORES enhances users' interactions with the recommender assistant, advancing the boundaries of AI-driven conversational interfaces. CORES offers multiple contextual representations that convey users' long-term shopping tendencies, enabling human-like and supportive conversations while enhancing user satisfaction and deep engagement. The system delivers a variety of visual responses, including VR visualizations, charts, and insights to create an immersive and informative shopping experience. By combining the capabilities of LLMs like GPT (generative pre-trained transformer) [15] and BERT (bidirectional encoder representations from transformers) [16], CORES provides informed feedback, emotional understanding, and rich contextual recommendations. It employs a hybrid embedding matching algorithm to recommend items by integrating multiple dimensions such as users' emotions, contextual data, item features, and geographical factors like shop location. The key contributions of our work can be summarized as follows:

1. We introduce CORES, an innovative decision support system specifically developed for virtual shopping scenarios. It can offer users customized recommendations for similar individual interests based on VR context, users' profiles, and users' preferences.

2. CORES uses users' prompts to detect emotions, analyzing both positive and negative emotions at specific times and locations to enhance the recommendation process.
3. We illustrate the functionalities of CORES in providing recommendations based on the most up-to-date technologies, integrating LLMs and RAG.
4. We conduct comparative experiments to demonstrate the usefulness and efficiency of our system. The results lead to higher levels of user satisfaction and reveal new models and directions for collaborative recommendation methods.

This article is structured as follows. Section 2 gives an overview of existing related work. Section 3 details the proposed methodology. Section 4 presents the developed prototype and its features. Section 5 presents the results obtained, and Section 6 concludes this paper and presents suggestions for future research.

2. Related Works

2.1. Virtual Reality-Based Recommendation System

In recent years, web-based or e-commerce recommendation systems have been widely used to increase sales by suggesting items to users. Virtual shopping is one of the most widely used e-commerce applications that allows users to interact with computer simulations to purchase and select virtual items.

The rapid advances in VR technology have brought new paradigms to traditional shopping designs and interactions [17,18]. VR provides a more effective dimension for meeting shopping needs. Compared with traditional in-store shopping, e-commerce has often been criticized for lacking hands-on interaction with products [19,20]. In addition, it can be difficult for users to gather relevant information when they are limited to two-dimensional model displays. However, customers and vendors can communicate more effectively through VR and overcome these obstacles [21–23]. For example, IKEA Kreativ [24] is a typical platform where customers can enter a virtual store and try virtual interactions. IKEA Kreativ aims to improve users' experiences with VR to support customers in their purchasing decisions.

Some studies show that VR can provide customers with more attractive, interactive, and immersive product information and thus improve the shopping experience [5,25]. Kim et al. [5] suggest that VR can serve as a social tool to enhance interactivity and improve cognitive abilities in the context of e-commerce. Considering these advantages, our goal is to utilize VR's comprehensive telepresence and rich interaction information to enhance the shopping experience. However, existing studies have overlooked the importance of personalized recommendations in VR, resulting in users feeling a lack of their needs and preferences being met. Our goal is to leverage VR technology to enhance the shopping experience, enabling users to explore products in an immersive environment.

2.2. Emotion-Driven Virtual Shopping

Emotion analysis within virtual environments improves the generalization capacities of emotions and interactions for decision making. An existing work examines how friendship strength can be integrated using Twitter as a source of large social networks (SNSs) [26]. Friendship strength is calculated using three categories of social data, including user-generated content, relationship information, and interaction information. In general, integrating social context into recommender systems significantly increases recommendation performance in terms of accuracy.

Other research has used social information (e.g., family, followers, friend list, relationships, and trusted and untrusted users) to improve prediction when providing recommendations. For example, in [27], Obata et al. proposed a rule-based recommendation system for mobile applications based on the user's contextual information. They used an emotion

model to define customized recommendations based on the customer's emotions. For the recommendation, they extracted emotions using the rating grid method to determine the positions of the emotions based on the extracted emotion model. Overall, emotional expressions and social interactions can integrate multimodal information for reviewing comments, which can consider various external factors such as VR and social context.

2.3. LLM-Based Virtual Shopping

At present, language processing is beyond the capabilities of most virtual shopping systems. However, recent advances indicate that this may soon change. As language processing continues to improve, virtual shopping environments will likely follow the trend and adopt language-based interactions. Given the potential of language processing to enhance the shopping experience, we believe it is an opportunity to investigate the use of LLM with recommendation strategies for virtual shopping. With LLM, a shopper could interact with a computer shopping agent, either through verbal communication or typed input, in a simulated natural dialogue. The agent must understand the request, seek clarification whether the request is vague or ambiguous, and successfully fulfill it. This process might involve retrieving and presenting information from a database, or in the case of a recommendation system, suggesting an item that best fits the shopper's needs.

Zhang et al. [28] emphasized the need to support users in VR environments and recommended products for them. They analyzed the visual features of VR environments using convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to ensure personalized recommendations based on users' preferences and previous interactions in VR environments. Other than 3D content representation of personalized VR experiences [29], other works dealing with encoding and processing methods included graph-based neural networks for efficient content recommendation systems. Chen et al. [30] used users' gaze direction, hand movements, and spatial navigation patterns to dynamically adapt content recommendations based on their current interests and engagement, thereby enhancing the overall VR experience. Authors in [31] demonstrated the effectiveness of hybrid recommendation approaches compared with popularity-based methods for recommending items in VR. In the domain of decision making, recent studies have begun to investigate the impact of additional contextual information on reasoning. For example, Lakehal et al. [32] achieved high-quality recommendations by automatically combining contextual rules.

Some researchers [33] have also integrated the contextual information not into users' profiles but into content-based recommendation systems in VR. These works do not link decision making to the LLM's ability to integrate contextual information and explain the results to the user. Therefore, we aim to ensure transparency by enabling the assistant to explain its results to the user while also considering the user's emotional behavior data, thereby enhancing its ability to provide personalized recommendations.

2.4. Limitations of Existing Approaches

Table 1 presents a review of related works based on LLM, context, social and emotion dimensions, and support for 3D visualization. The existing recommendation approaches aim to combine different recommendation approaches to achieve consistently improved recommendations. However, these approaches neglect emotional information and the shopping context in VR during the recommendation process. In the proposed VR environment, the user is represented as an avatar, allowing them to interact with the environment and other avatars. This type of shopping simulation is highly suitable for research, as it allows us to observe conversations between the user and the assistant in a shopping-focused context. Furthermore, there are currently no methods for shopping assistants that are specifically tailored for shopping simulations. Considering virtual reality dimensions and

social collaboration platforms with emotional ratings during the recommendation process can improve the shopping experience and increase recommendation accuracy.

Table 1. Related works comparisons.

Related Works		LLM	Context	Social	Emotions	3D Vis.
VR-Based Recommendation	Jiang et al. [17]	×	×	×	×	✓
	Suh et al. [18]	×	×	×	×	✓
	Han et al. [19]	×	×	×	×	✓
	Jessen et al. [20]	×	×	×	×	✓
	Kowalczyk et al. [21]	×	×	×	×	✓
	Meißner et al. [22]	×	×	×	×	✓
	Wu et al. [23]	×	×	×	×	✓
	IKEA Kreative [24]	×	×	×	×	✓
Emotion-Based VR Shopping	Lee et al. [25]	×	✓	✓	×	✓
	SNSs [26]	×	✓	✓	✓	×
	Obata et al. [27]	×	✓	✓	✓	×
	Zhang et al. [28]	✓	✓	×	×	✓
LLM-Driven VR Shopping	Wang et al. [29]	✓	✓	×	×	✓
	Chen et al. [30]	✓	×	×	×	✓
	Sharma et al. [31]	✓	×	×	×	✓
	Lakehal et al. [32]	×	✓	×	×	×
	Kim et al. [33]	×	×	×	×	✓
Chen et al. [34]		✓	×	×	✓	×
Our Proposed Approach		✓	✓	✓	✓	✓

3. The Proposed Conversational System: CORES

To enhance user experience, we investigate the performance of LLMs [15] in fine-tuning domain-specific e-commerce models to generate embeddings for textual item descriptions and find relevant items that match a user's query. Furthermore, we use RAG [35] to generate personalized recommendations, particularly by integrating VR, based on the retrieved items, user's context, user's emotional rating, user's history conversations, and user's preferences.

We consider a VR shop that represents various virtual visualized items from different stores, each specialized in selling different types of products such as kitchen appliances, sporting goods, etc. When a person wants to buy a product, the most difficult task is selecting the right product that meets their needs, preferences, and contextual requirements. This is where the conversational recommender comes into play. In addition, the conversational recommender should ensure customization based on the user's context and emotional ratings. Therefore, we provide additional details of the recommended item, which are generated based on GPT-4 [36–38]. These additional details are provided as a list of the item's characteristics (e.g., color, materials, size, delivery time, etc.). In the following, we present the proposed framework, including the architecture, context dimensions, and functionalities of CORES.

3.1. Problem Formulation and Overview

We formulate the problem of this study as follows:

1. The VR shop consists of several digital stores $DS = \{DS_1, DS_2, \dots, DS_M\}$. Each digital store DS_i is specialized in selling specific genres of items (kitchen, sports, choices, etc.). We define each store DS_i by the spatial 3D coordinates x_i , y_i , and z_i (latitude, longitude, and height) (e.g., latitude: 40.7128° N, longitude: 74.0060° W, height: 10 m).
2. The item dataset $D = \{I_1, I_2, \dots, I_N\}$ is classified by digital store and genre, where N represents the total number of items.

3. Each item I_i is identified by id , where it has a description d and a set of features $F_{ij} = \{f_{i1}, f_{i2}, \dots, f_{iM}\}$. M represents the total features by item. These features could include attributes such as size, price, color, brand, style, material, performance, popularity, durability, and other additional attributes for specific product genres such as laptops defined by battery life and capacity.
4. The conversational recommendation handler must recommend n items in the form of answers $A_i = \{(I_1, E_1, V_1, S_1), \dots, (I_n, E_n, V_n, S_n)\}$ for the user based on its i th query Q_i , where I_j represents the j^{th} recommended items, E_j represents the corresponding j^{th} explanation, V_j represents the corresponding j^{th} VR visualization, and S_j represents the corresponding j^{th} insight.
5. The user query Q_i is defined in terms of genre and intentions (i.e., characteristics). For example, if a user asks for kitchen appliance recommendations or mentions specific genres, this indicates their intentions, including modern styles and durability to discover new kitchen appliances. We consider explicit and implicit users' preferences expressed during interactions with CORES, for example, a user looking for a portable device with a good battery life for gaming and work, with implicit preferences from the conversation history.
6. We have L comments from N users for each item I_i using social accounts.

CORES take as inputs (1) the user's request, (2) the user's preferences, (3) historical interactions, (4) the user's context, (5) stores DB, (4) items DB, and (6) the social registry. Its final output is the recommended items with explanations, visualizations, and insights. The overall workflow is described in detail in the next section.

3.2. General Architecture of CORES

The architecture of the proposed system is illustrated in Figure 1. The architecture consists of the following modules:

1. **VR shopping app:** It is designed to handle conversations between users and the system, known as the front end. It is responsible for capturing the user's inputs and visualizing the responses in the form of 3D items through the VR headset based on the Unreal Engine platform.
2. **CORES:** It is known as the back end of the system, responsible for receiving and processing users' information. It includes the following modules:
 - **Conversation state management (CSM):** It is responsible for identifying users' interests and extracting users' preferences, context, and emotions from users' inputs and incorporating them with the response generator module to generate responses. CSM is based on LLM, Langchain (<https://www.langchain.com/>, accessed on 11 February 2025). This Python library integrates GPT-4 models for effectively assisting users during the shopping virtual experience by applying text embedding techniques and answering questions about recommended items. CSM also uses GPT-4 models to analyze users' prompts and determine the tasks to be sent to the recommendation engine and response generator. CSM tracks the user's emotional state, preferences, and interactions, which informs the recommendation engine to offer personalized suggestions based on the user's emotions.
 - **Recommendation engine:** It is responsible for the computation of matching scores based on the user's attention, preferences, item genre, location, and current time. It performs the attention-based context-aware emotion-driven recommendation process to match the query with the item dataset to find the list of relevant items. These are then sent to the response generator module to generate two mapped vectors, including explanations and insights. The recommendation

engine gives suggestions based on the user's emotions, shaping the conversation. If the user feels satisfied or dissatisfied, the system updates the conversation to improve future recommendations.

- **Response generator:** This module uses the items list generated by the recommendation engine as inputs to generate recommendation responses in terms of explanations and insights. Afterwards, these responses are then transferred to the VR loader module for visualization through the VR headset. The response generator includes the following modules:
 - **Recommendation explication generator:** It generates recommendation explanations based on the results of attention matching, social matching, and context matching. Then, it formulates the matching results with GPT-4 as recommendation explanations and sends them to the VR loader for visualization.
 - **Recommendation insight generator:** It uses GPT-4 to generate insights based on data collected from internal items datasets and external data sources. The recommendation insight generator responds by analyzing and transforming data into valuable information from customer feedback and purchase history to find popular products and trends for guiding the decision process.
- **VR loader:** It sets up the VR environment by configuring VR-specific components such as cameras, controllers, and interaction mechanisms. Our VR loader module is based on the Unreal Engine platform, which offers robust support for handling and visualizing photorealistic real-world objects. The VR loader includes the following modules:
 - **Scene loader:** it generates and sets up the VR scene, which will serve as a container for visualizing recommended items.
 - **3D model loader:** It loads the 3D model of the recommended items list into the scene. It also adjusts the position and scale of the loaded items within the scene to fit the user's visualization needs.
 - **Explication and insight loader:** it loads text explanations and insights into the generated scene after adjusting the text properties such as position, rotation, scaling, font, color, etc.
- **Registries and Profiles:** They represent the knowledge base of CORES and collect data about users, items, and conversation histories of specific shopping domains. These data are used to generate responses for users represented in recommended items list, insights, and explanations. To address privacy concerns related to data collection, we have integrated the context-aware attribute-based access control (Cx-CP-ABE) [39] approach mainly based on blockchain technology [40]. This method effectively tackles privacy issues in systems like CORES, particularly when handling sensitive data such as personal or shopping information. With Cx-CP-ABE, access to sensitive data is restricted only to the authorized users, depending on the context of their request. This enhances both privacy and security by granting access only to data that are relevant to the user's role or specific needs. Granular controls, such as considering users' preferences or session details, allow for a more personalized and privacy-conscious experience while maintaining robust data protection.

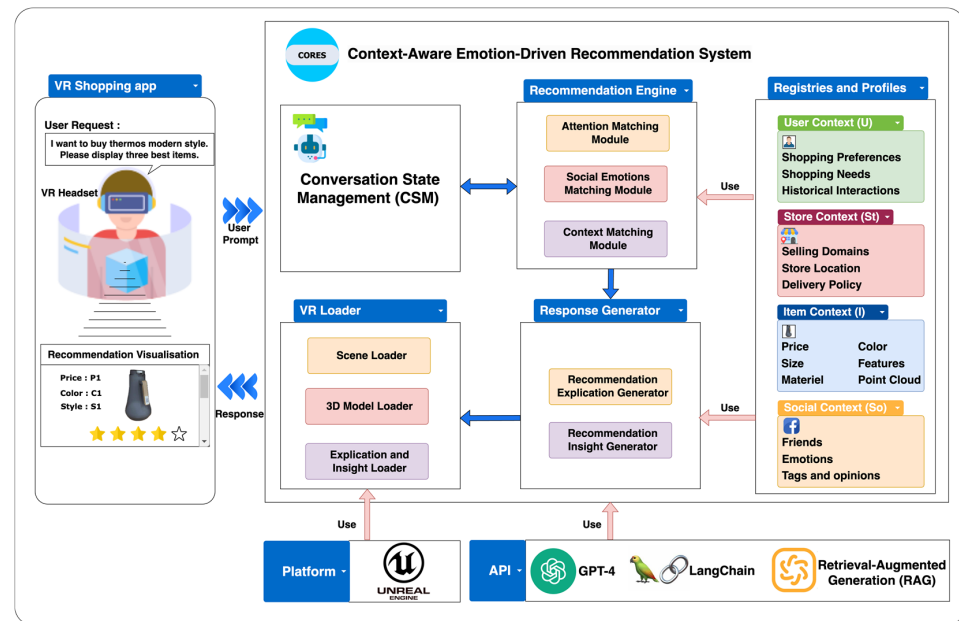


Figure 1. The general architecture of CORES using VR.

3.3. The Context Dimensions in CORES

In CORES, context is categorized into four key areas: *User*, *Item*, *Store*, and *Social*, as depicted in Figure 2. To offer relevant recommendations, CORES analyzes users' requests to understand their intentions and shopping preferences, while also considering their emotions and environmental factors, such as available items and delivery times. The system incorporates several profiles and registers to effectively manage and process this contextual information.

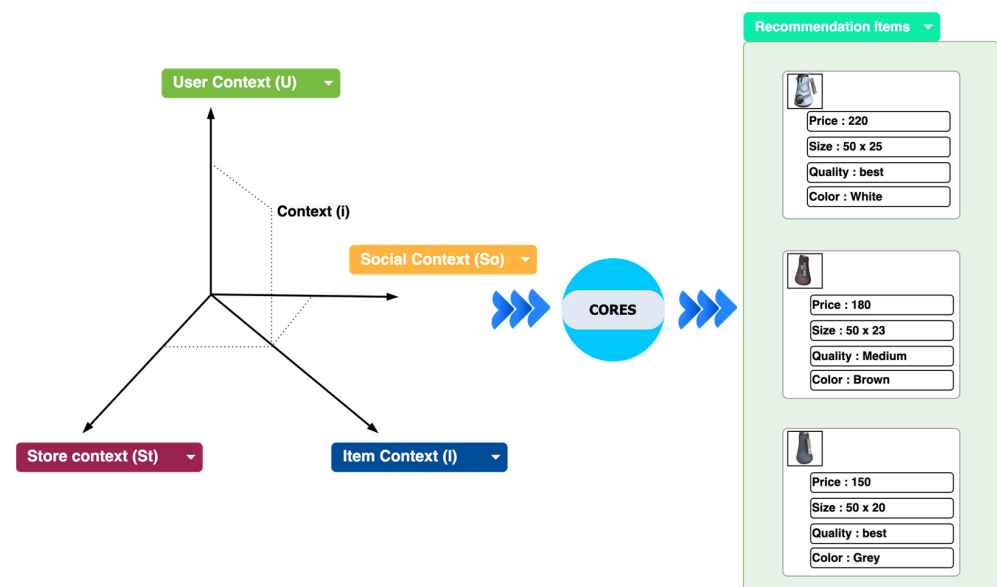


Figure 2. The context dimensions in CORES.

- User context:** A successful recommendation system must address users' diverse and evolving needs by considering both their current and long-term preferences. CORES achieves this by analyzing users' shopping needs, social conversation histories through interactive queries, historical interactions (e.g., users' past purchases), and explicit preferences (e.g., product type, size, color) to analyze past social interactions and conversations using emotion-driven AI models, like RoBERTa [41,42], and contextual

factors such as delivery location and time via GPS to consider users' proximity to stores, delivery locations, or preferred delivery times. By integrating preferences like item durability, quality, price, services, and style, CORES ensures personalized recommendations that enhance user satisfaction.

- **Store context:** it consists of context information related to each store, highlighting key features such as selling domains, store location, delivery policy, etc.
- **Item context:** Each item in CORES includes rich contextual details like price, color, size, and material, along with highly detailed 3D models for realistic visual representation. The 3D reconstruction process involves data capture using LIDAR and cameras to scan items from multiple angles, generating a point cloud for detailed modeling. MeshLab refines and textures the resulting 3D mesh, which is exported in OBJ format for visualization in Unreal Engine, enhancing the immersive VR experience (see Figure 3).

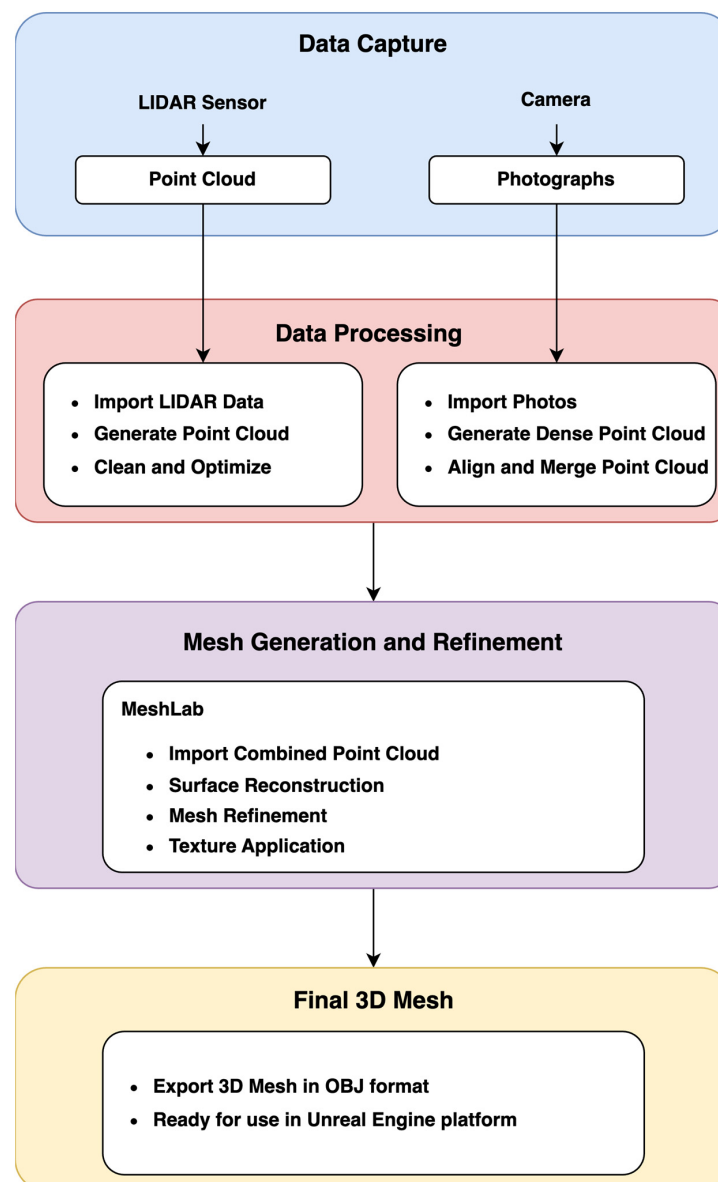


Figure 3. Three-dimensional mesh generation.

- **Social context:** CORES integrates the user's social media data, such as Facebook activity, to incorporate friends' emotions and interactions with items. Recommendations consider factors like price, quality, and social history, promoting customer-centric e-commerce. By analyzing emotional expressions such as admiration, anger, and

gratitude, CORES tailors recommendations to provide both immediate and long-term satisfaction. This social dimension ensures personalized emotionally relevant suggestions, aligning with user satisfaction and fostering long-term success in e-commerce.

3.4. The Emotional Detection Mechanism

In CORES, emotion detection is performed by advanced NLP and emotion classification techniques. The system uses transformer-based models, specifically RoBERTa (a variant of BERT) [41,42], which are fine-tuned for emotion classification. This approach allows CORES to interpret both the semantic meanings and the emotional states in users' inputs. As shown in Figure 4, the process of the emotion detection mechanism involves the following steps:

1. **Tokenization:** Users' comments on items are divided into tokens (words or sub-words). This step helps manage unfamiliar words by splitting them into smaller interpretable units.
2. **Embedding:** After tokenization, the RoBERTa model converts the tokens into embeddings. These embeddings map words with similar meanings or contexts to nearby locations in a vector space, enabling the model to capture deeper semantic relationships.
3. **Fine-tuning the RoBERTa model:** The pre-trained embeddings are refined during fine-tuning to enhance the model's ability to classify emotions. This optimization is based on patterns observed in the training data.
4. **Emotion classification:** RoBERTa generates context-aware embeddings for each token. These embeddings are processed to produce a classification vector, where each element corresponds to a specific emotion category (e.g., happiness, joy, fear, anger).
5. **Inference and prediction:** When a new user input is received, it performs tokenization, and embeddings are created for each token. These embeddings are passed through the fine-tuned model to predict the emotion, which predicts the emotion by assigning a confidence score to each emotion category. The emotion with the highest score is selected as the predicted label.

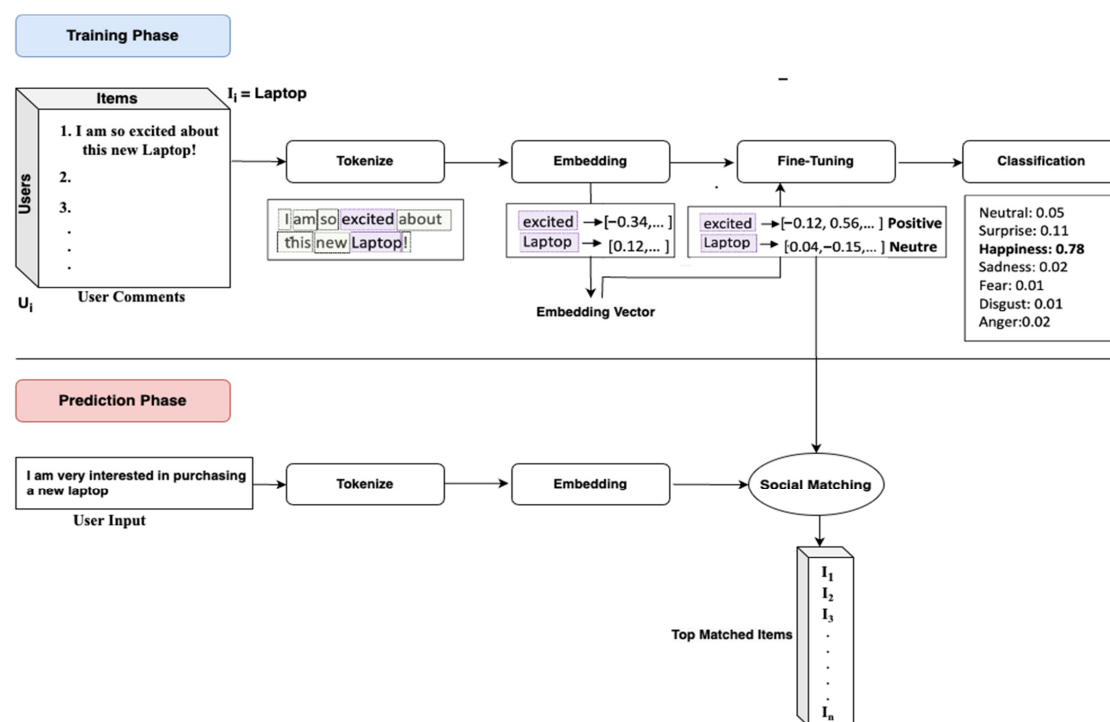


Figure 4. The emotional detection mechanism.

This process enables CORES to accurately detect and classify emotions during real-time user interactions.

3.5. Workflow of CORES

To develop CORES, we utilize a hybrid embedding approach powered by LLM tools to ensure robust and personalized recommendations. The workflow consists of seven key steps as shown in Figure 5 as follows: (1) extracting features and creating item embeddings, (2) creating store embeddings, (3) creating emotion embeddings, (4) creating users' request embedding and modeling preferences, (5) performing hybrid embedding-based matching, (6) conducting hybrid embedding-based ranking, and (7) generating RAG-based responses.

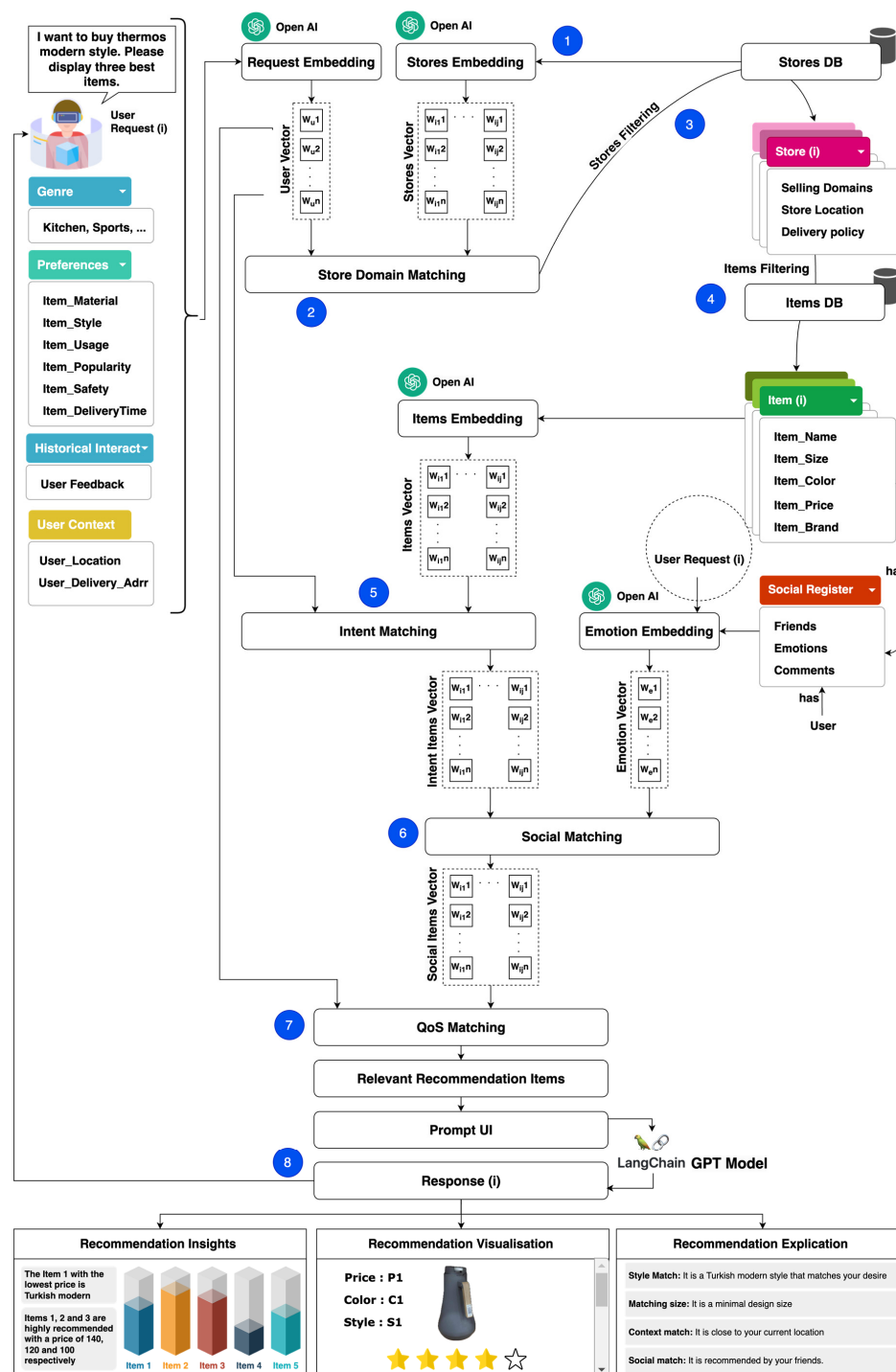


Figure 5. The workflow of the item recommendation process.

3.5.1. Item Embedding

The first phase of CORES focuses on creating vector representations of items based on their textual descriptions in the dataset to identify retrieval for the user. Using the GPT-4 model and sentence transformers, the system computes embedding weights for each item's description, encapsulating its key features. This process involves the following:

- (1) **Feature extraction:** decomposing each item description d_i into words or sub-words f_{j,d_i} , where j is the index of each feature within the description and i is the number of items varied from 1 to N .
- (2) **Feature embedding:** assigning a unique embedding vector $E_{f_{j,d_i}}$ to each feature f_{j,d_i} using GPT-4, capturing its contextual meaning of the whole sequence.
- (3) **Concatenating embeddings:** combining all feature embedding vectors of an item into a single vector E_{d_i} , as defined by:

$$E_{d_i} = \text{Concat}(E_{f_{1,d_i}}, E_{f_{2,d_i}}, \dots, E_{f_{M,d_i}}) \quad (1)$$

where M is the total number of features in the item description d_i (with $i \in [1..N]$).

- (4) Repeat all steps 1–3 for the next item description d_i .

3.5.2. Store Embedding

The second phase focuses on processing store locations and their associated attributes to generate store embedding vectors. Each store is linked to a specific shopping activity, and its delivery address and store address are converted into GPS coordinates to estimate delivery time. These details are then passed to the recommendation engine as a store embedding vector. The embedding vector for each store DS_i , E_{DS_i} is constructed using a similar process to item embedding and is defined as:

$$E_{DS_i} = \text{Concat}(E_{\text{dom}_{DS_i}}, E_{\text{loc}_{DS_i}}, E_{\text{police}_{DS_i}}) \quad (2)$$

where $E_{\text{dom}_{DS_1}}, E_{\text{loc}_{DS_2}}, E_{\text{police}_{DS_3}}$ represent the embedding vectors for the selling domain, the store's location, and the delivery policy of the digital store DS_i , respectively. By following the same process, the embedding vectors of all digital stores are obtained.

3.5.3. Emotion Embedding

The third phase of CORES integrates an innovative emotional component using the RoBERTa model [41,42]. This advanced NLP-based model identifies and processes the emotional sentiments expressed by users in their input messages. RoBERTa is chosen for emotion detection in CORES due to its superior performance in NLP tasks, particularly in emotion analysis and classification. We have specifically incorporated RoBERTa because it is fine-tuned to recognize a wide range of emotions (e.g., happiness, anger, sadness, excitement), which is crucial to accurately interpreting users' emotions. For each emotional comment c_{ij} posted by a user U_j on an item I_i , the RoBERTa model extracts feature embeddings. The comment c_{ij} consists of a sequence of words $word_1.word_2 \dots word_{M_{ij}}$. The model processes the comment and extracts the final layer embeddings to create an emotional comment embedding $E_{c_{ij}}$ defined as:

$$E_{c_{ij}} = \text{Concat}(E_{word_{1_{ij}}}, E_{word_{2_{ij}}}, \dots, E_{word_{M_{ij}}}) \quad (3)$$

where M_{ij} is the total number of words in the comment c_{ij} . By following the same embedding process with RoBERTa, the embedding vectors of the corresponding comments under the same item are obtained.

3.5.4. User's Request Embedding and Preferences Modeling

We focus on transforming a user's request Q into a contextual embedding that incorporates a user's genre, needs, preferences, and current context, facilitating the recommendation of relevant and context-specific items. This is performed with the help of GPT-4 embeddings and transformers:

(1) **Automatic tokenization of a user request to generate feature embedding**

The user's request Q is split into words Qf_i . Each feature Qf_i is mapped to an embedding vector E_{Qf_i} using GTP-4 embeddings.

(2) **Concatenation of feature embedding vectors to produce user's request embedding**

The embedding vectors of all features are concatenated to obtain the final embedding vector for a user's request E_Q :

$$E_Q = \text{Concat}(E_{Qf_1}, E_{Qf_2}, \dots, E_{Qf_M}) \quad (4)$$

where M is the total number of features in the user's request Q . The embedding vector E_Q represents a user's request, its features, and its semantic meaning.

(3) **User's preferences modeling**

CORES evaluate how closely items match the user's specific preferences by analyzing various features, historical interactions, and the user's current context. The proposed approach defines weights to these dimensions W_{feature} , W_{hist} , W_{ctx} . The user's preference weighting vector W_{pref} is defined as:

$$W_{\text{pref}} = \text{Concat}(W_{\text{feature}}, W_{\text{hist}}, W_{\text{ctx}}) \quad (5)$$

3.5.5. Hybrid Contextual Embedding-Based Search

We instruct CORES to recommend relevant items that match a user's preferences and context through a hybrid embedding-based matching and ranking process. CORES combines four matching measures, as follows:

(1) **Domain Embedding Matching**

The business domain similarity Sim_Domain measures the similarity between the user's preferred genre E_{Q_genre} and the store's selling domain $E_{\text{dom}_{DS_i}}$ of the digital store DS_i using cosine similarity. The objective is to assess whether the types of items available in the store match the user's interest, calculated as follows:

$$\text{Sim_Domain}(Q, DS_i) = \frac{E_{Q_genre} \cdot E_{\text{dom}_{DS_i}}}{\|E_{Q_genre}\| \cdot \|E_{\text{dom}_{DS_i}}\|} \quad (6)$$

Stores with the highest similarity scores proceed to the next step.

(2) **Feature Embedding Matching Based on User's Preferences**

The feature embedding similarity score Sim_Feature between the user's query Q and the item description d_i from the previously filtered store is calculated using a weighted cosine similarity formula as follows:

$$\text{Sim_Feature}(Q, d_i) = \sum_{j=1}^M w_j \cdot \frac{E_{Qf_j} \cdot E_{f_j, d_i}}{\|E_{Qf_j}\| \cdot \|E_{f_j, d_i}\|} \quad (7)$$

where w_j is the weight for feature f_j , and E_{Qf_j} and E_{f_j, d_i} are the embeddings for the user's and item's feature, respectively.

Items with the highest similarity scores are filtered for the next step.

(3) Social Embedding Matching Driven by User's Emotions

This step evaluates the emotional alignment between a user's query and comments on filtered items. Each comment's embedding vector $E_{c_{ij}}$ is compared with the user's query emotion embedding vector $E_{Q_emotion}$ using cosine similarity $Sim_Emotion$ as follows:

$$Sim_Emotion(Q, c_{ij}) = \frac{E_{Q_emotion} \cdot E_{c_{ij}}}{\|E_{Q_emotion}\| \cdot \|E_{c_{ij}}\|} \quad (8)$$

The similarity scores of all comments for an item are aggregated to compute the overall social emotion-based similarity Sim_Social :

$$Sim_Social(Q, d_i) = \frac{1}{L} \sum_{j=1}^L Sim_Emotion(Q, c_{ij}) \quad (9)$$

Items with comments that strongly match the user's emotional context receive higher scores and are prioritized for the next step. This process ensures recommendations that consider both the user's emotional intent and social feedback.

(4) QoS Matching Based on User's Context

The store's location and the user's delivery address are crucial factors in calculating the delivery time for the recommended item from the prior step. The delivery time is estimated using the Euclidean distance between the store and the delivery address, divided by the average speed of the delivery vehicle. Additional elements, such as transportation mode, road conditions, and real-time traffic, can be factored in to enhance the accuracy of the travel time estimate. The context-aware QoS similarity score (QoS) reflects this estimated delivery time using the following formula:

$$QoS(DS_i, U) = DT = \frac{D_i}{speed} \quad (10)$$

where DT is the estimated delivery time, D_i is the distance between the digital store DS_i and the user's delivery address, and $speed$ is the average speed of the delivery vehicle. The Euclidean distance D between the digital store and the delivery address is given by:

$$D = \sqrt{(x_{DS_i} - x_d)^2 + (y_{DS_i} - y_d)^2} \quad (11)$$

where (x_{DS_i}, y_{DS_i}) are the GPS coordinates of the digital store DS_i and (x_d, y_d) are the GPS coordinates of the delivery address. Items from stores with the shortest delivery times and highest positive emotional feedback proceed to the next stage.

3.5.6. Hybrid Contextual Embedding-Based Ranking

The final recommendation score is calculated using a weighted linear combination of feature similarity, social similarity, and QoS similarity scores.

$$Similarity = \alpha \times Sim_Feature + \beta \times Sim_Social + (1 - \alpha - \beta) \times QoS \quad (12)$$

where α and β are the weight indicating the importance of each criterion (features, social, QoS, delivery). Users visually choose their preferred features, social factors, and QoS using GUI of VRShop. This selection process aids in calculating the weights for α and β by assigning a high weight to the most preferred feature, a low weight to the least preferred, and a neutral weight to unselected options. Preferences vary among users; some

prioritize items that match their specific preferences over social feedback, while others value proximity and high social or emotional ratings. Leveraging feedback from others is essential to enhancing the success of product recommendations. Items are ranked based on their final scores, with the highest-ranking ones recommended to the user. Items from stores with the shortest delivery times and highest user attention are prioritized for the next step. The proposed item recommendation is shown in Algorithm 1.

Algorithm 1 Items Recommendation Algorithm	
Inputs:	Q : User Query; d : Items descriptions; D : digital stores, C : comments W_f : features weights; α, β : similarity weights; U_{ctx} : User context, n : number of top items
Outputs:	Top Recommended Items
Begin	
(1):	Foreach digital store $DS_i \in DS$ Do
(2):	$DomainScore \leftarrow Sim_Domain(Q, DS_i)$
(3):	IF ($DomainScore == 1$) Then
(4):	$D \leftarrow D \cup (d_j \in DS_i)$;
(5):	End
(7):	Foreach item $d_i \in D$ Do
(8):	$FeatureScore \leftarrow Sim_Feature(Q, d_i)$
(9):	Foreach comment $c_{ij} \in C$ Do
(10):	$EmotionalScore \leftarrow EmotionalScore + Sim_Emotion(Q, c_{ij})$
(11):	$SocialScore \leftarrow EmotionalScore / L$
(12):	$QoSscore \leftarrow QoS(U_{ctx}, DS_i)$
(13):	$Score \leftarrow \alpha \times FeatureScore + \beta \times SocialScore + (1 - \alpha - \beta) \times QoSscore$
(14):	End
(15):	End
(16):	$SortScores \leftarrow$ Sort the items in D by their Score in descending order
(17):	$TopItems \leftarrow$ take the first n items from $SortScores$
(18):	Return $TopItems$
End	

3.5.7. Response Generation Process in CORES

The response generation process, as outlined in Algorithm 2, involves the use of GPT-4 models. This process mainly consists of the following steps:

(1) Semantic Analysis of User's Inputs

When the conversational recommender agent receives the identified emotions from RoBERTa and the user's query, it leverages a GPT-4 model that includes the conversation prompt template and historical conversations. The goal of this step is to classify the user's intent, extract relevant entities, and derive the underlying meaning of the user's input.

(2) Determining the Answer Based on the Extracted Intents

The recommendation engine works with the response generator to assess whether the user's input has an existing response or requires additional information. If the intent corresponds to a final result, the system stores this result in the conversation history and provides a recommendation explanation to the user's VR. If the intent is new or requires an updated response, the process moves to the next step.

(3) Response Generation based on Matching Results

In this step, the system generates responses by combining three components: the response generator, the recommendation explanation, and the recommendation's VR visualization and insights. GPT-4 processes the user's request using identified entities (such as tokens and emotions from Step 1), and applying the appropriate response format. CORES provides detailed explanations for recommendations, clarifying aspects such as item features (e.g., colors, sizes), emotions, and context (e.g., store location, delivery time). The generated responses can be visualized in 3D VR. Additionally, CORES utilizes GPT-4's capabilities with chart generation tools to provide new insights that better meet the user's needs. The process is driven by the input prompt *Prompt_1*, and further details can be found in Appendix A.

Algorithm 2: Response Generation

Inputs: *Q*: User Query.
Pr_C: Conversational Recommender Template.
Pr_E: Explication Generation Template.
Pr_V: Visualization Generation Template.
Pr_I: Insights Generation Template.
GPT_f: GPT_4 Generation function.
HI: conversations history;

Outputs: *R*: Response Results;

Begin

- (1): $R \leftarrow \emptyset$
- (2): **Foreach** $Q^i \in \text{user queries}$ **Do**
- (3): $I, E \leftarrow \text{GPT_4}(Q^i, Pr_C, HI, R)$
- (4): **IF** (*I* is Final Answer) **Then**
- (5): $HI \leftarrow \text{Prompt_4}(Q^i, Pr_C, E)$
- (6): **return** *E*;
- (7): **END IF**
- (8): **IF** (*I* is Explication Generation) **Then**
- (9): $R1 \leftarrow \text{Prompt_1}(Q^i, Pr_C, Pr_E, E)$
- (10): $R \leftarrow R \cup R1$
- (11): **END IF**
- (12): **IF** (*I* is VR Visualization Generation) **Then**
- (13): $R2 \leftarrow \text{Prompt_4}(Q^i, Pr_C, Pr_V, E)$
- (14): $R \leftarrow R \cup R2$
- (15): **END IF**
- (16): **IF** (*I* is Insight Generation) **Then**
- (17): $R3 \leftarrow \text{Prompt_1}(Q^i, Pr_C, Pr_I, E)$
- (18): $R \leftarrow R \cup R3$
- (19): **END IF**
- (20): **End FOR**
- (21): **Return** *R*

End

3.5.8. Three-Dimensional Scene Reconstruction Workflow in CORES

Figure 6 depicts the 3D scene reconstruction process of CORES, designed to improve the user's experience by combining visual features, interactivity, and contextual information. The process begins with the scene loader, which creates the virtual space where items will

be displayed. At the same time, CORES sends a list of recommended items to the 3D model loader. The 3D model loader then imports the recommended items in OBJ format into the VR scene. Unreal Engine enhances these objects with rich textures and vibrant colors, ensuring precise and realistic visualization within the environment. Additionally, Unreal Engine sets up the scene by adjusting the scale and optimizing the appearance and performance of the 3D objects in the VR space, enabling real-time user interactions such as zooming and rotating. The explication and insight loader is used to display product descriptions and provide textual explanations, offering additional context and insights about the visualized items. Once all components are integrated, the VR environment is fully populated with the recommended items, enabling users to seamlessly select and interact with them.

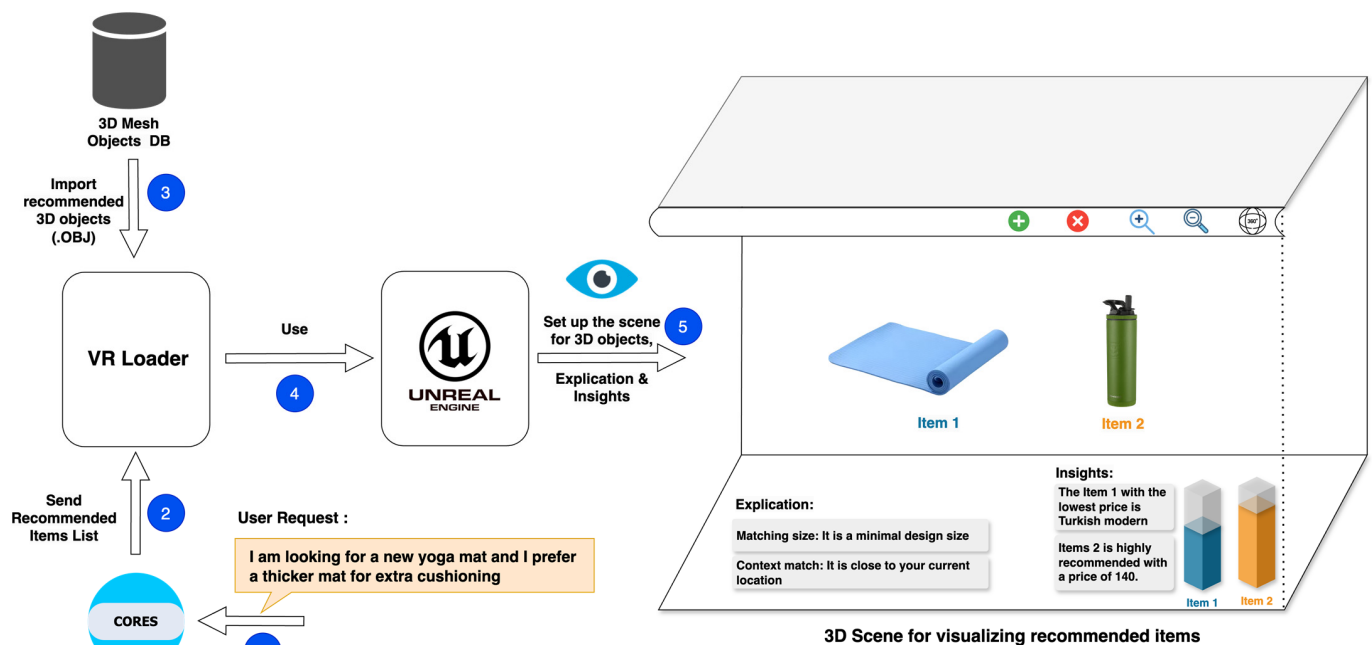


Figure 6. Three-dimensional scene reconstruction workflow.

4. Implementation and User Scenario

4.1. Implementation

CORES for VRShop was developed to offer users immersive shopping experiences across various stores. The CORES VRShop app (1.0) is designed to be compatible with VR headsets, but it does not require installation directly on the headset; instead, it can be run in conjunction with other devices, such as a PC, laptop, or smartphone, which handle the computational and data processing tasks. However, VR headsets provide a fully immersive 3D virtual environment and enable natural interactions through motion tracking, hand gestures, and head movement. By integrating a recommendation system, CORES enhances the experience by suggesting products based on users' preferences and behavior. Users can enter VRShop with a VR headset, navigating through a virtual mall that contains stores showcasing 3D replicas of physical products. Each store displays high-definition models of products, allowing users to interact with them by picking them up, rotating, and zooming in to examine intricate details. Users can access additional information such as price, color, size, material, product reviews, and application demonstrations.

CORES analyzes users' behaviors, preferences, and past purchases to recommend relevant products. Personalized suggestions are provided based on browsing histories, search queries, and interaction patterns. For instance, if a user frequently searches for

fitness equipment, the system will prioritize related items like yoga mats, dumbbells, and activewear.

The CORES system is implemented consisting of two parts: the front end (VR users' interface) and the back end (CORES recommender). The front end processes users' requests by cleaning and formatting them (removing unnecessary characters), while the back end provides assistance and product recommendations. The front-end prototype utilizes 3D scene reconstruction techniques (as described in Section 4), with a Python library (<https://streamlit.io/>, accessed on 11 February 2025) linking the user's interface to VR capabilities, enabling the placement of 3D objects within a virtual store. On the back end, the Langchain library facilitates the integration of GPT-4, providing tools to create accurate prompts for improved responses from GPT-4 and other large language models (LLMs).

CORES is demonstrated through its ability to answer users' queries and recommend relevant products based on the user's preferences. The system adapts to context changes (e.g., size, style) to filter and refine item quality, ensuring better personalized recommendations. The user's first interaction with VR shopping is shown in the welcome interface (Figure 7), while digital stores are illustrated in Figure 8.

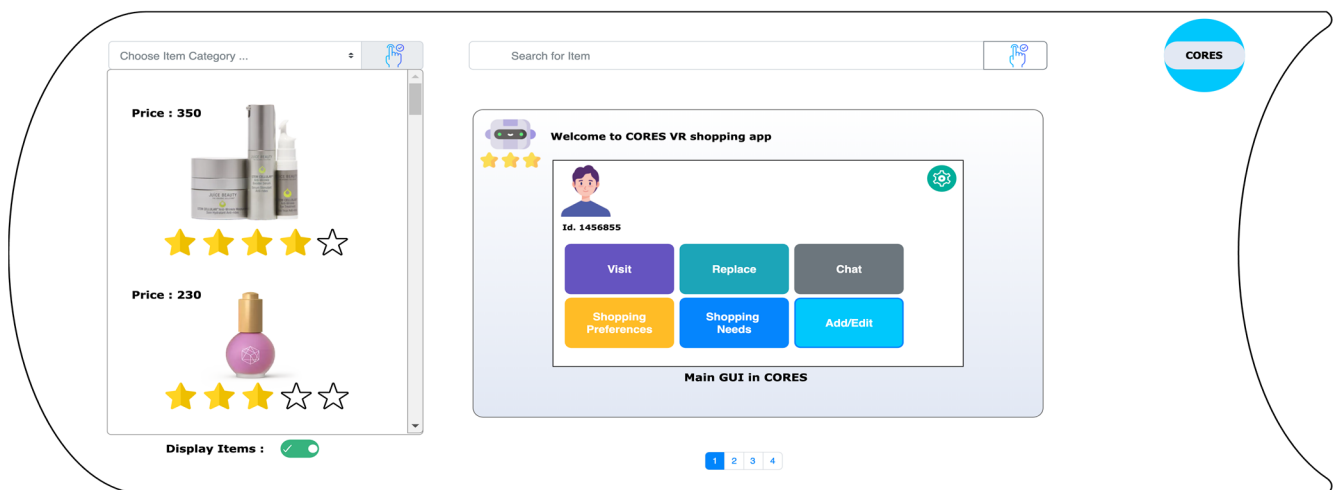


Figure 7. VRShop welcome interface.

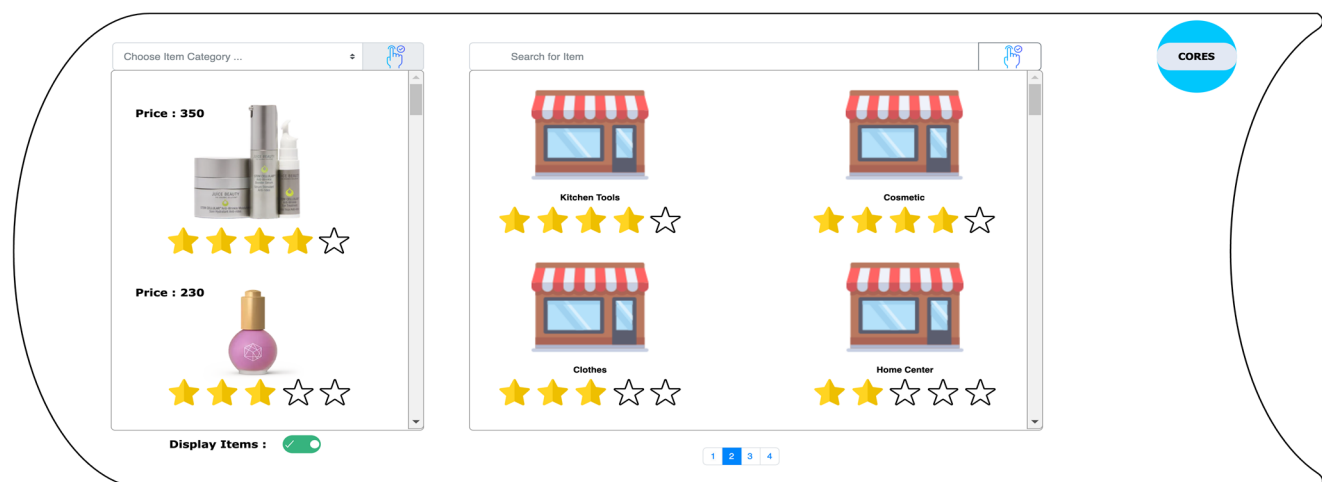


Figure 8. Digital stores of VRShop.

4.2. Functional Details

The context-aware emotion-driven recommendation mechanism for items is the core feature of our prototype, designed to enhance the quality of item recommendation. The prototype provides the following key functionalities:

1. Interactive capabilities and usage history.
2. Top products' display according to the user's request.
3. Dynamic items' addition in the VR environment.
4. Profile-based item categorization.
5. Update management and customized recommendation.
6. Social network feedback integration.
7. Rating and performance reports in terms of accuracy and delivery time.

4.3. User Scenario: Adam's Virtual Shopping Experience

The CORES VRShop offers a dynamic shopping experience, demonstrated through various specialized stores. In this scenario, Adam, a fitness enthusiast, uses his VR headset to enter VRShop, where he finds himself in a lively virtual mall with numerous stores, each dedicated to different product categories.

Adam chooses to explore a sporting goods store. He browses products such as running shoes, exercise equipment, and sportswear, using the realistic 3D models to closely inspect the items, providing him with a tactile shopping experience. However, after a while of searching, Adam does not find what he wants. He decides to use the system's conversational interface to find a specific product. A user interface prompts him to select a product category and specify his preferences, where he inputs: *"I'm looking for a new yoga mat and I prefer a thicker mat for extra cushioning, and I like vibrant colors. Eco-friendly materials would be a plus"*.

The VRShop assistant immediately processes these preferences and suggests three personalized options: a 6 mm thick eco-friendly yoga mat in teal, a 5 mm thick non-slip mat in purple, and a 7 mm thick high-density mat in green. Adam examines these suggestions in 3D, rotating the mats to assess their texture and thickness. He repeats this with the purple and green mats, as depicted in Figure 9.

Figure 9. Specification of user's preferences.

Each item is equipped with four main features: update, insights, comments, and chat. For example, if Adam wants to learn more about the teal yoga mat, he can click the

“Insight” button to view detailed information about its features, displayed on the user interface as shown in Figure 10. Additionally, he can check real-time reviews from other users by clicking on the product image, providing him with insights from social networks, as illustrated in Figure 11. Appendix A provides more detailed prompts in CORES.

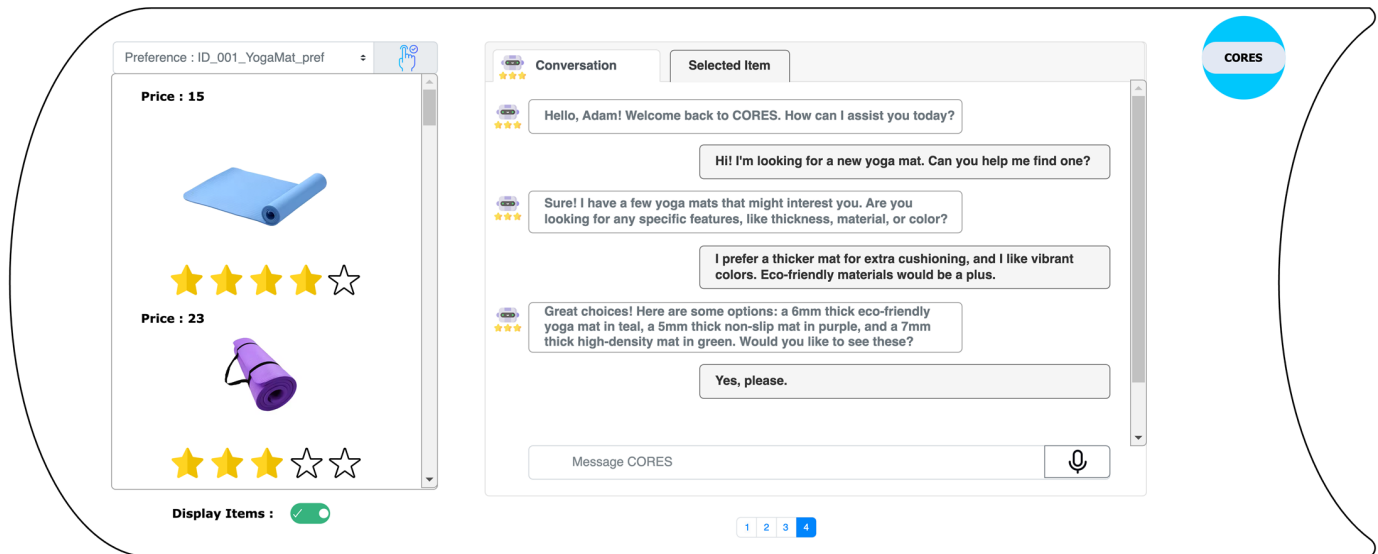


Figure 10. Selection of virtual shop.

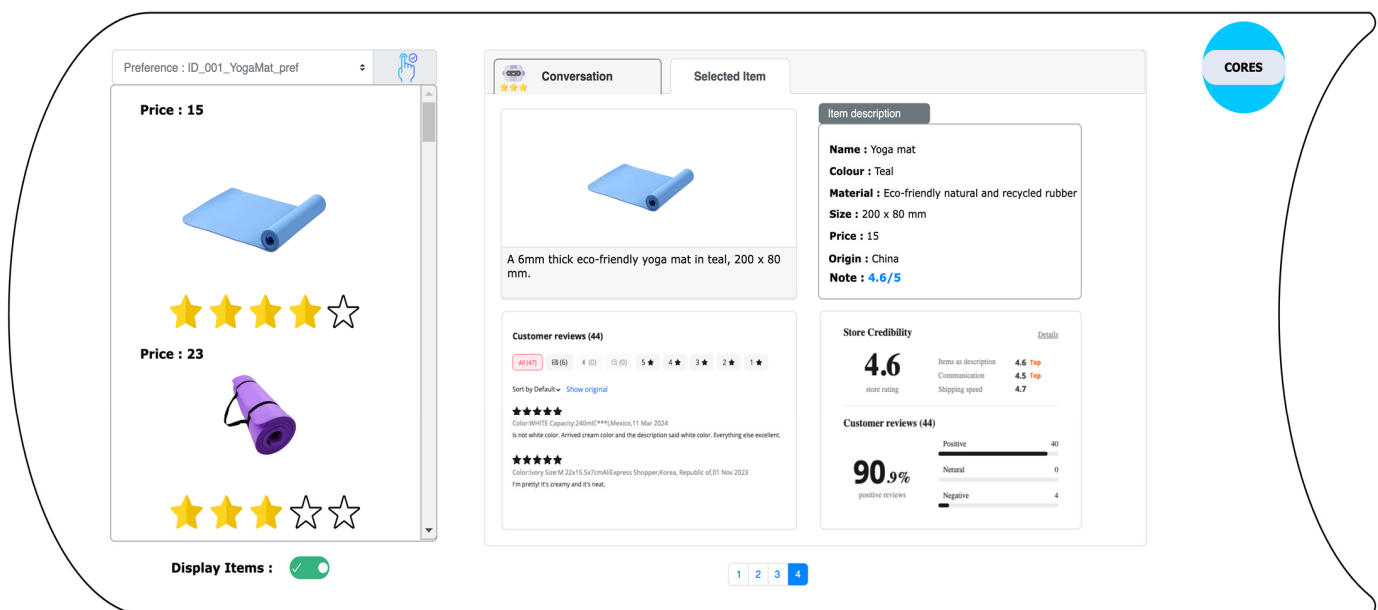


Figure 11. Display matching results with relevant comments.

Another feature allows Adam to click the “Chat” button, which opens a dialogue interface for voice and text communication, enabling him to ask more questions or receive further suggestions. If Adam wants to act on a recommendation, he can click on the corresponding comment to view it in real time.

Figure 12 illustrates the intermediate results of each step in the reconstruction workflow. First, the VR loader receives recommended items from CORES and constructs a digital representation of the 3D scene. Next, Unreal Engine generates virtual 3D models of the recommended items, adding realistic surface details. Finally, the restored VR scene is created by aligning the 3D models of the items and incorporating explications and insights.

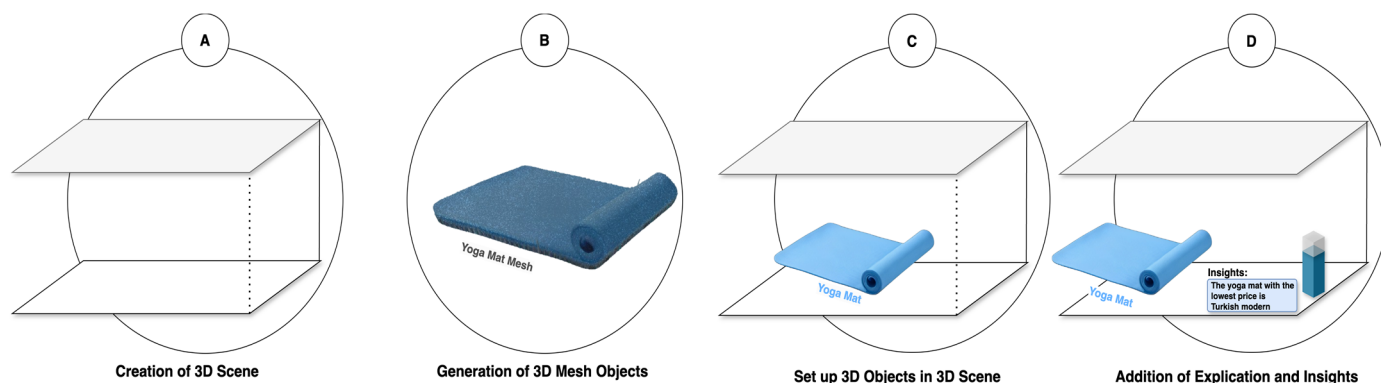


Figure 12. Some intermediate results of scene reconstruction.

4.4. Evaluation

4.4.1. Experimental Questions and Evaluation Metrics

Our experiments aim to evaluate the efficiency and effectiveness of CORES as a VR-based decision-making tool, exploring utility in everyday life. Specifically, we focus on the following two experimental questions:

Q1: how does the contextual information provided by VR affect the performance of CORES and the decision-making behavior of users?

Q2: how do different users' emotions affect the quality of CORES' recommendations?

To address these questions, we evaluate CORES' performance based on the following metrics:

- **Accuracy:** the proportion of correct answers among all answers produced by CORES.
- **Precision (Pre):** the proportion of relevant and correct answers among all the answers generated by CORES.
- **Recall (Rec):** the ratio of correctly identified answers to the total number of expected correct answers.
- **F1–Score:** a weighted average of precision and recall, providing a balanced measure of CORES' performance.

$$Accuracy = \frac{\#correct\ answers}{\#all\ answers\ of\ CORES} \quad (13)$$

$$Pre = \frac{\#relevant\ and\ correct\ answers}{\#all\ answers\ of\ CORES} \quad (14)$$

$$Rec = \frac{\#correct\ answers}{\#expected\ answers\ of\ CORES} \quad (15)$$

$$F1 - Score = 2 \times \frac{Pre \times Rec}{Pre + Rec} \quad (16)$$

4.4.2. Datasets

This study uses four different datasets, each centered on a specific product category: kitchen tools, home furniture, yoga mat, and cosmetics. A brief description of each dataset is provided below:

- (1) **Kitchen tools dataset:** this dataset contains information about kitchen tools, including attributes such as color, size, brand, category, and materials used.
- (2) **Home furniture dataset:** This dataset provides information on various furniture items for different spaces (living rooms, bedrooms, offices). Each item contains in-

formation such as ID, style, dimension (length, width), type, material, price, and experts'/vendors' assigned tags.

- (3) **Yoga mat dataset:** this dataset focuses on a range of yoga mats with attributes such as name, brand, material (e.g., PVC, TPE, cork), thickness, dimensions (length and width), colors, eco-friendliness, customers' ratings, and durability.
- (4) **Cosmetics dataset:** This dataset covers a variety of beauty products, including skincare, makeup, haircare, and fragrances. Each product entry includes the name, brand, category, ingredients, volume, and price.

4.4.3. Question Collection

As part of the study, questions related to these datasets are collected, classified, and aligned with CORES' objectives, enabling us to answer requests regarding kitchen appliances and provide detailed explanations, visualizations, and insights. A total of approximately 30 questions were gathered and categorized into 3 groups (as shown in Table 2).

Table 2. Types of questions in CORES.

Types of Questions	#Questions	Examples of Requests
Find items	10	I want to buy a thicker mat for extra cushioning, vibrant colors. Please display three best items. Find the number of mixed styles. List all thicker mats with low prices.
VR visualization	10	Provide details of more products. I want to show products in 3D vision.
Insight analysis	10	Draw a bar chat for top purchased products. Are there any prices we can increase, and by how much, without losing customers?

4.4.4. Parameter Settings

During the evaluation, a test dataset is used, consisting of tuples that include 30 responses related to the found articles, VR, and insights. The prepared responses include the following: (1) a user's preference question to recommend an item from the dataset, (2) a correct target answer/context related to the e-commerce domain, and (3) the answer generated by CORES in response to the question. The yoga mat dataset used in this experiment consists of 300 items within the e-commerce system. This dataset plays a crucial role in evaluating the efficiency of the proposed CORES system, as outlined in Table 3. CORES stands out from traditional methods by integrating context-aware AI, emotion-driven recommendations, and immersive VR shopping experiences. Unlike static ratings, CORES analyzes a wide range of factors, including users' preferences, emotional insights, interaction histories, and contextual attributes (e.g., *style, size, and color*), to provide personalized and dynamic recommendations. To manage new or unique items, CORES incorporates the following: (1) cold start handling that matches the attributes of new items to users' preferences using content-based filtering and (2) predicts emotional patterns based on similar items, ensuring accurate and reliable evaluations of previously unseen products.

Table 3. Experimental setup.

Parameter	Value
#items list	300
#relevant items	20
# non-relevant items	270
#users	100
# emotions for evaluation	150,000
#positive emotions	83,075
#negative emotions	66,925
Preferences W_{color} , W_{style} , $W_{eco-friendless}$	0.5, 0.3, 0.2
α , β	0.4, 0.2

4.4.5. Participants and Data Collection

The system is designed to help the user to find suitable items by leveraging emotional data collected using Facebook’s RoBERTa model. This dataset contains over 150,000 comments, with 2 main classes of emotions (e.g., *positive* and *negative*) divided into 7 types of emotions (e.g., *neutral*, *surprise*, *happiness*, *sadness*, *fear*, *disgust*, *anger*). Table 4 illustrates sample results showcasing users’ satisfaction levels for a specific product. To gather data, users’ feedback samples are randomly selected, and items are distributed across three stores for evaluation. Table 4 highlights five key fields, including age groups (in intervals of 10 years), positive and negative ratings based on product price and quality, and corresponding percentages of satisfaction or dissatisfaction. When a product achieves a satisfaction score of 70% exceeding the expert-defined threshold of 60%, the system automatically displays a happy emoticon, signaling a recommendation to the user for that product. This approach combines users’ feedback and emotional cues to enhance the recommendation process.

Table 4. Emotions regarding yoga mat item (excitement, satisfaction, disagree).

Age Groups	Age	Positive	Negative	Positive (%)	Negative (%)
Middle	[20–30]	8	5	0.615	0.385
	[31–40]	12	1	0.923	0.077
	[41–50]	8	4	0.667	0.333
Elderly	[41–50]	7	5	0.583	0.417
	[61–70]	10	3	0.769	0.231

5. Results

To assess the efficiency and effectiveness of VRShop, we employed multiple evaluation metrics. All experiments were conducted on a system with an Intel Core i3-5005U CPU (2.00 GHz), 4 GB RAM, running Windows 8 (64-bit).

5.1. Evaluation of Efficiency

Table 5 presents a performance comparison of the recommendation system using various embedding methods applied to the yoga mat dataset. The evaluation involved different item counts, ranging from 50 to 300.

Table 5. Performance comparisons of VRShop for different embedding methods.

Method	Dataset Size	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
BERT	50	76.0	90.9	66.7	76.9
	100	76.0	90.9	66.7	76.9
	150	80.0	90.0	75.0	81.8
	200	81.0	93.3	78.6	89.4
	250	80.8	82.4	97.0	85.1
	300	76.3	90.9	66.7	76.9
TF-IDF	50	98.0	100	98.0	99.0
	100	96.0	100	96.0	98.0
	150	96.7	100	96.7	98.3
	200	97.0	100	97.0	98.5
	250	95.6	100	95.6	97.8
	300	97.3	100	97.3	98.6

Table 5. Cont.

Method	Dataset Size	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
TF-WordVect	50	94.0	100	91.4	95.7
	100	87.0	94.3	83.3	83.3
	150	92.7	96.9	92.9	94.9
	200	90.5	94.2	92.9	93.5
	250	92.0	100	88.6	94.3
	300	92.0	100	88.6	94.3
Proposed	50	97.0	75.0	100	85.7
	100	96.0	71.0	100	83.7
	150	96.7	72.2	100	83.9
	200	95.5	66.7	100	80.0
	250	95.6	70.3	100	82.5
	300	96.7	70.6	100	82.8

5.2. Evaluation of CORES With and Without Emotional Aspects

To evaluate the performance of the CORES system with and without emotional aspects, we conducted a 36 h test to collect sufficient data. The test showed a statistically significant difference in user satisfaction, with a p -value of 0.001. Participants were divided into two groups. Group A interacted with a traditional recommendation engine that did not consider emotions, while Group B used CORES, which integrated emotional insights into its recommendations.

Figure 13 shows a comparison of CORES' performance with and without emotional integration, focusing on key metrics such as users' satisfaction, retention rates, emotional responses, and recommendation accuracy. The results clearly demonstrate that CORES, with emotional integration, enhances recommendation accuracy and improves users' satisfaction, engagement, and retention. These findings highlight the value of emotional intelligence in enhancing online shopping experiences.

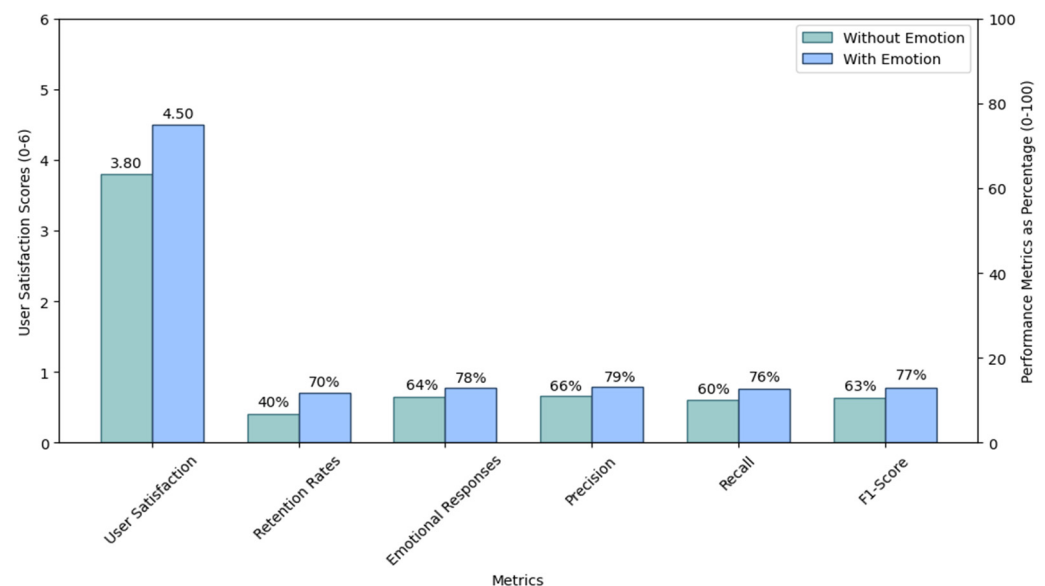


Figure 13. Performance comparison with/without emotion in VR. User satisfaction scores (0–6) vs. performance metrics as percentages (0–100).

Unlike traditional shopping apps like Amazon, which mainly rely on past behaviors such as purchase histories, CORES offers a personalized recommendation system by considering the user's current emotional state. It dynamically adjusts its recommendations

based on emotional context, time, and location. Moreover, CORES integrates social context by adapting recommendations according to the emotional states of the user's friends. By continuously analyzing users' emotions and monitoring interactions, CORES fine-tunes the shopping experience in real time to enhance users' satisfaction and engagement. This customization includes comprehensive product explanation, detailed material descriptions, emotional feedback, prompt responses to inquiries, and reminder emails with transparent pricing to effectively address users' concerns.

5.3. Evaluation of Average Accuracy With and Without Context and Emotions

Further experiments were conducted to evaluate the system's performance when incorporating emotions, preferences, and context. Table 6 highlights the results for item recommendations using the yoga mat dataset. The findings reveal that the highest accuracy, 97%, was achieved when combining context with preferences. When emotions were also included, the accuracy increased to 100%. These results demonstrate that integrating emotions with context and preferences significantly enhances the system's performance compared with relying solely on context and preferences.

Table 6. Performance evaluation of the recommendation system with and without context and emotion consideration selection (preference color >> thickness >> eco-friendly).

Performance Metrics (%)	Without Context	With Context and Preferences	With Context, Preferences, and Emotions
Accuracy	94%	97%	97%
Precision	75%	75%	75%
Recall	60%	100%	100%
F1-score	66.7%	85.7%	85.7%

5.4. Influence of the CORES Modules on System Performance

To evaluate the impact of the CORES modules including conversation state management (CSM), recommendation engine, response generator, VR loader, and registries and profiles on overall system performance and user decision making, we used K-fold cross-validation ($K = 5$) to assess key performance metrics, including accuracy, precision, recall, and F1-score. The results are illustrated in Figure 14, providing a comprehensive analysis of each module's contribution to enhancing system effectiveness. Furthermore, to validate the statistical significance of these modules' impact, we applied the Friedman test to analyze the effect of each module. This analysis was supported by calculating chi-square (χ^2) and p -value, based on the same set of subjects to test the following hypotheses:

H₀: the removal of the module does not significantly impact the system performance.

H₁: the removal of the module significantly impacts the system performance.

For the CSM module, the Friedman test exhibited statistically significant differences in the results ($\chi^2 = 18.24$, $p = 0.002$). Cross-validation revealed that the inclusion of CSM had a substantial impact on CORES' recommendations, with accuracy ($94.2\% \pm 1.3\%$), precision ($72.1\% \pm 1.5\%$), recall ($98.0\% \pm 1.2\%$), and F1-score ($83.4\% \pm 1.7\%$) all being significantly affected. These results highlight the pivotal role of the CSM module in tracking users' states and interaction histories, ensuring more accurate context-aware recommendations.

Regarding the recommendation engine, the Friedman test demonstrated highly statistically significance differences in the results ($\chi^2 = 22.35$, $p = 0.0005$). Cross-validation revealed that the recommendation engine impacted the system performance, particularly in recall ($91.6\% \pm 2.3\%$) and F1-score ($78.7\% \pm 2.1\%$) compared with accuracy ($92.3\% \pm 1.5\%$) and precision ($70.2\% \pm 1.7\%$). This suggests that richer contextual information and social emotions improved CORES' precision, leading to more relevant recommendations. This

also supports the concept that users benefited from context-aware social feedback, thus increasing their satisfaction.

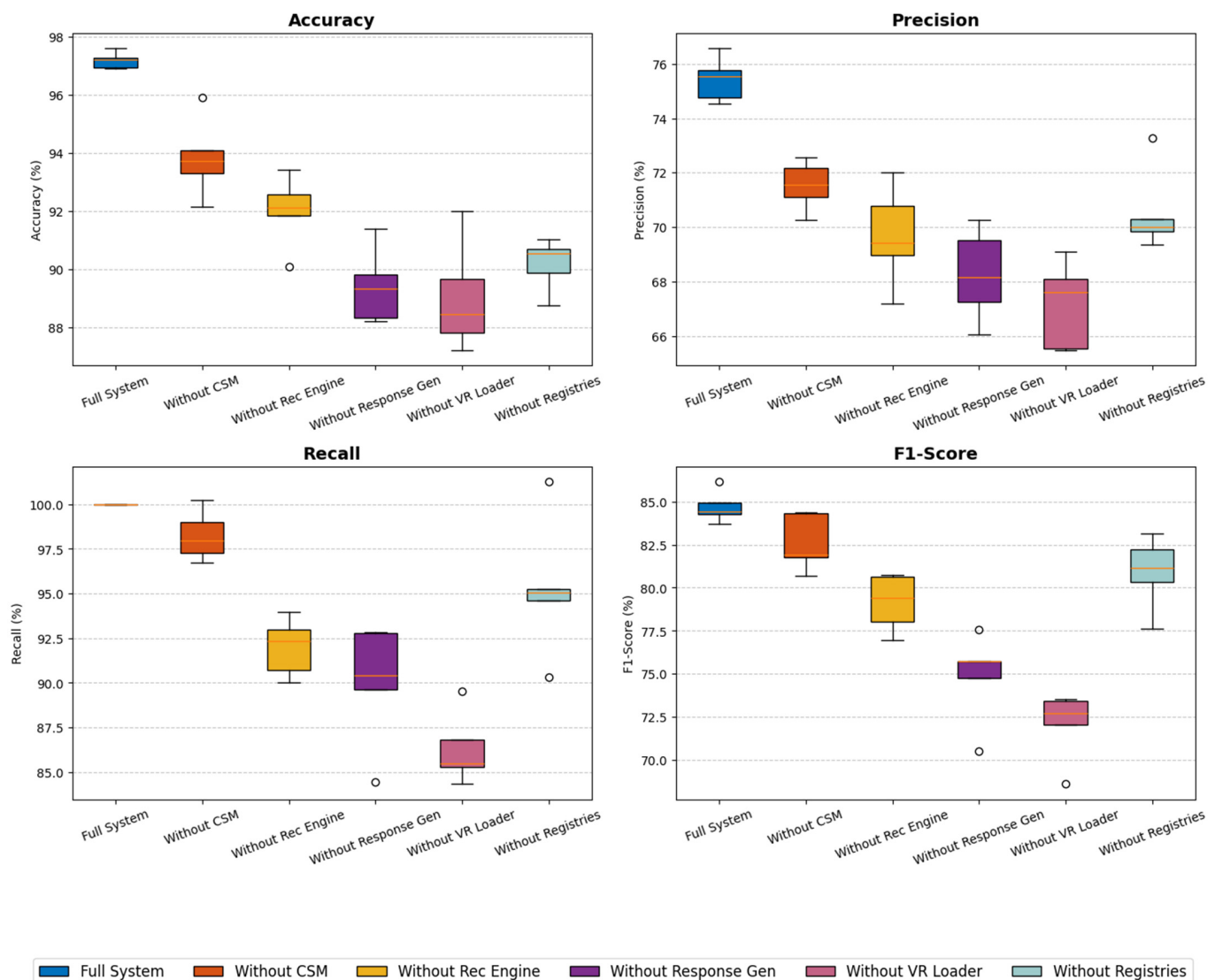


Figure 14. The performance results of CORES without individual modules; p -values < 0.05 and 0.01 , respectively, indicating statistical significance.

Furthermore, cross-validation also revealed a significant impact on the system's performance due to the response generator, with accuracy ($90.1\% \pm 1.6\%$), precision ($67.4\% \pm 2.1\%$), recall ($89.7\% \pm 2.0\%$), and F1-score ($75.5\% \pm 2.5\%$) all being affected. Furthermore, the Friedman test also showed statistically significant differences in the results ($\chi^2 = 19.50$, $p = 0.001$). These results suggest that semantic analysis and intent extraction enhanced the credibility of explanations, making recommendations clearer and more transparent. Participants found that categorizing explanations by dimension improved clarity and user-friendliness.

For the VR loader, cross-validation demonstrated a significant statistical impact on the system performance, with accuracy ($88.9\% \pm 2.1\%$), precision ($66.8\% \pm 2.5\%$), recall ($86.5\% \pm 3.1\%$), and F1-score ($72.7\% \pm 2.8\%$). The Friedman test revealed statistically significant differences in the results ($\chi^2 = 16.78$, $p = 0.004$). These results show that VR integration enabled CORES to provide more precise item recommendations based on contextual information. However, the absence of VR-based visual interactions reduced users' engagement, leading to lower recall rates.

For registries and profiles, the Friedman test indicated statistically significant differences in the results ($\chi^2 = 19.50$, $p = 0.001$). Cross-validation demonstrated that updating profiles and registries improved accuracy ($91.3\% \pm 1.8\%$) and precision ($69.5\% \pm 2.0\%$). This shows that incomplete registries and profiles led to increases in redundant or irrelevant responses, reducing overall system relevance.

The results show that all modules had statistically significant impacts on the system performance. The recommendation engine and the VR loader had the most substantial impact on recall and F1-score. The response generator was critical for enhancing precision, while CSM ensured system coherence and accuracy. Overall, CORES enabled users to gain comprehensive insights through interactive visually immersive product information, enhancing their decision making and improving their overall experiences.

5.5. Evaluation of Effectiveness

The effectiveness of the proposed recommendation system was evaluated using the yoga mat dataset, incorporating context, preferences, and emotion ratings. Table 7 compares the performance of this method against other techniques.

Table 7. Response time comparison with/without context and emotion in VR.

Method	Embedding Vector Size	Memory Space	Execution Time (s)
BERT	768	$768 \times (n + 1)$	24.688
TF-IDF	50	$50 \times (n + 1)$	0.0355
TF-WordVect	50	$50 \times (n + 1)$	0.2182
Our Proposed	768	$768 \times (n + 1)$	0.3267

The proposed system achieved an average response time of 3267 ms, slightly outperforming methods such as BERT and TF-WordVect, even with the use of a high-dimensional 768-dimensional embedding vector. In contrast, the TF-IDF-based recommendation system achieved a much shorter execution time of 355 ms due to its reduced embedding vector size of 50. By leveraging personalized recommendation based on users' context, ratings, interaction histories, and preferences and VR features such as store location and item availability, the proposed system demonstrated an ability to optimize computational efforts while maintaining high effectiveness.

5.6. Discussion

Context is the key to most decisions in all areas of business. In today's competitive marketplace, many contextual factors make purchasing decisions complex, dynamic, and expensive. However, most recommendation systems rely on simplistic methods that are limited to manually provided users' preferences and needs.

To address these challenges, e-commerce must invest in VR technologies to handle complex and conflicting decisions. Our research showed that combining VR with advanced AI can effectively automate and control the online shopping process. Additionally, combining the VR context with social emotions enhances decision-making accuracy, achieving a 95.6% success rate. The CORES approach further enhances this process by delivering faster response times compared with conventional methods.

During users' interviews, we found that integrating participants' experiences greatly enhanced decision making. To optimize this process, it is crucial to adopt an iterative user-centered approach with feedback loops. While CORES can independently assist users in their decision making, its effectiveness is increased significantly when it leverages collective knowledge and prior experiences of a wider user base. This approach enables CORES to identify patterns, preferences, and best practices that extend beyond individual cases, establishing a strong foundation of contextual knowledge. Moreover, this contextual

information helps users to understand the rationale behind the recommendations. Our study focused on a case involving shopping items within four VR contexts: size, style, color, and material. However, this approach may not be possible to easily transfer to other product categories without adaptations, such as including additional contextual factors like multiple product types and characteristics. Exploring these factors could offer new research opportunities.

Despite the effectiveness of the CORES system in item recommendation, there are several limitations as follows that need to be addressed to enhance system performance and user experience:

- (1) **Refining technical details:** Implementing more advanced simulation methods could enhance system fidelity and increase automation, providing a more realistic and immersive user experience. This improvement could also enable a more dynamic interaction with virtual objects, such as real-time modifications of textures and colors, which are key elements in designing VR environments.
- (2) **Capturing dynamic in-app behavioral data:** The current reliance on social network data for recommendation generation may limit the system adaptability to different user behavior patterns. Our evaluation results showed that approximately 40% of participants with Facebook friends were actively expressing emotions and interacting with items. Nevertheless, the system maintained high accuracy due to its hybrid approach, which integrated historical interactions, content-based items, contextual information, time and location factors, and user feedback. These findings suggest that while social network data can provide valuable insights, they may not fully capture the engagement level of all users, limiting the system's effectiveness in real-world scenarios. We intend to explore in-app social interactions as a more reliable and adaptable alternative, enhancing the system's ability to well understand user behavior more accurately and build greater trust.
- (3) **Integrating deep learning and real-time analytics:** Although the system effectively integrates context-aware models, further optimizations are required to ensure scalability for large-scale VR deployments. Integrating deep learning and real-time analytics capabilities will improve system performance and responsiveness, enabling seamless interaction with dynamic user behaviors and adapting efficiently to environmental changes.
- (4) **Recognizing facial emotions in VR:** To enhance emotional depth and immersion, we plan to integrate embedded cameras and external sensors for real-time facial expression tracking. This includes analyzing facial muscle activity such as eyebrow movement and lip dynamics to capture subtle emotional changes and dynamically adjust VR environments and interactions based on detected emotions.

Future enhancements to CORES would benefit from incorporating historical insights and intelligent decision models to recognize recurring behaviors and trends. This would help create a more realistic and personalized experience for users. Enhancements could also enable real-time interactions, such as modifying the textures and colors of virtual items, which play a critical role in online shopping. Furthermore, continuous adaptation is a vital area for future development. By integrating insights from human–computer interactions and cognitive psychology, CORES can dynamically evolve through real-time user interactions within the VRShop. This would ensure that CORES remains responsive to changing contexts and user behaviors, further enriching the overall shopping experience.

6. Conclusions

Today, AI is an extremely important tool for businesses. Virtual assistance offers a tangible opportunity for innovation, as companies opt to provide an additional service to their

customers while optimizing processes such as recommendation support and insight analysis. Within this framework, we have developed a novel system called the context-aware emotion-driven recommendation system (CORES), which combines different technologies and approaches to ensure more satisfying performances by considering VR, emotional understanding, and contextual knowledge within LLM. The CORES system is structured into two parts: the VR user and LLM. LLM utilizes advanced prompt engineering, including RAG. CORES is implemented in the e-commerce domain as a useful and efficient tool in the context of customized products. The main goal of CORES is to understand users' emotions and to respond accordingly in a dynamic interaction mode, based on the user and shopping context. As for CORES' performance, the results obtained from experiments conducted on various datasets showed that CORES successfully recommended the right products aligned with users' preferences, achieving high levels of accuracy and efficiency on widely recognized public item datasets. Future research will explore the impact of real-time distributed users' interfaces with a broader range of emotional categories in e-commerce systems, while providing customers with more advanced interaction modalities.

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Appendix A

Table A1. PROMPT details in CORES.

CORES VRShop Assistant: "Hello, Adam! Welcome back to CORES. How can I assist you today?"
Adam: "Hi! I'm looking for a new yoga mat. Can you help me find one?"
CORES VRShop Assistant: "Sure! I have a few yoga mats that might interest you. Are you looking for any specific features, like thickness, material, or color?"
Adam: "I prefer a thicker mat for extra cushioning, and I like vibrant colors. Eco-friendly materials would be a plus."
CORES VRShop Assistant: "Great choices! Here are some options: a 6 mm thick eco-friendly yoga mat in teal, a 5 mm thick non-slip mat in purple, and a 7 mm thick high-density mat in green. Would you like to see these?"
Adam: "Yes, please."
CORES VRShop Assistant: "I've placed the yoga mats in front of you. You can pick them up and take a closer look."
<i>Adam picks up the teal yoga mat and examines it in 3D, rotating it to see the texture and thickness. He repeats this with the purple and green mats.</i>
Adam: "I like the teal one. Can you give me more insights about its features?"
VRShop Assistant: "Certainly! The teal yoga mat is made from eco-friendly TPE material, 6 mm thick for optimal cushioning, and has a non-slip surface for better grip during workouts. It's also lightweight and easy to carry."
Adam: "I'll take the teal yoga mat."
VRShop Assistant: "Excellent choice! I've added the items to your cart. Would you like to proceed to checkout?"
Adam: "Yes, let's do that."
VRShop Assistant: "Your total is \$45.99. Would you like to use your saved payment method?"
Adam: "Yes, that works."
VRShop Assistant: "Your order is confirmed, Adam. Your item will be delivered to your address within 3–5 business days. Is there anything else I can assist you with today?"
Adam: "No, that's all. Thank you!"
VRShop Assistant: "You're welcome! Have a great day and enjoy your new yoga mat!"

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